

| Question number | Answer                                                                                                                                                                                                                                                                                                           | Mark |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 4(a)(i)         | <p>The correct answer is <b>C (forensic entomology)</b></p> <p><i>A is incorrect because chromatography is a separation method</i><br/> <i>B is incorrect because dendrochronology is the study of tree growth rings</i><br/> <i>D is incorrect because heterozygosity is a measure of genetic diversity</i></p> | (1)  |

| Question number | Answer                                                                                                                                                                                                                                                                                      | Mark |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 4(a)(ii)        | <p>The correct answer is <b>D (1, 2 and 3)</b></p> <p><i>A is incorrect because all three factors provide the information</i><br/> <i>B is incorrect because all three factors provide the information</i><br/> <i>C is incorrect because all three factors provide the information</i></p> | (1)  |

| Question number | Answer                                                                                                                                                                                                                                                                                                                                          | Mark |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 4(b)(i)         | <p>The correct answer is <b>B (pushing a temperature probe into the liver)</b></p> <p><i>A is incorrect because this would measure external temperature of body + air temperature effect</i><br/> <i>C is incorrect because the mouth is not deep inside the body</i><br/> <i>D is incorrect because the arm pit is not inside the body</i></p> | (1)  |

| Question number | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Mark |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 4(b)(ii)        | <p>The correct answer is <b>A (increase, increase)</b></p> <p><i><b>B</b> is incorrect because more clothing will keep the body warmer and increase the time estimate + higher ambient temperature would reduce heat loss and increase the estimate</i></p> <p><i><b>C</b> is incorrect because more clothing will keep the body warmer and increase the time estimate + higher ambient temperature would reduce heat loss and increase the estimate</i></p> <p><i><b>D</b> is incorrect because more clothing will keep the body warmer and increase the time estimate + higher ambient temperature would reduce heat loss and increase the estimate</i></p> | (1)  |

| Question number | Answer                                                                                                                                                                                                                                                                                                                                                                         | Additional guidance                                                                                | Mark |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|------|
| 4(c)(i)         | <p>An explanation that includes the following points:</p> <ul style="list-style-type: none"> <li>ATP decreases as {oxygen levels / aerobic respiration (by muscle cells)} decrease (1)</li> <li>lactic acid increases as anaerobic respiration (in muscle cells) takes place (1)</li> <li>pH decreases because of the {production of / increase in} lactic acid (1)</li> </ul> | <p><b>ACCEPT</b> used during muscle relaxation</p> <p>drop in pH reduces ATP synthase activity</p> | (3)  |

| Question number | Answer                                                                                                                                                                                                                                                                                                                                                                                                  | Additional guidance                                                                                                                                                                                                                                                                                                                                                                     | Mark |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 4(c)(ii)        | <p>An explanation that includes three of the following points:</p> <ul style="list-style-type: none"> <li>because muscle contraction increases and decreases (with time after death) (1)</li> <li>gives two different estimates (1)</li> <li>(degree of) muscle contraction affected by {other factors / named factor} (1)</li> <li>judging (extent of) muscle contraction is subjective (1)</li> </ul> | <p><b>ACCEPT</b> stiffen and relax<br/>idea that muscles will be relaxed before and after contraction</p> <p><b>ACCEPT</b> two quoted times that extent of contraction is the same<br/>only useful between 2.4 and 36 hours after death<br/>cannot estimate the TOD after 36 hours as the muscles will be relaxed<br/>similar contraction at {flat region / between 15 to 23 hours}</p> | (3)  |

